Company name: Ambient Smart Homes

Category: Fitness / Wellness

Summary: Ambient makes products for healthy homes.

Description

Ambient plans to make a smart home product that improves indoor air quality. Our health depends so much on the quality of the air that we breathe, yet very few of us know what is in the air around us. Most smart home products focus on two things: managing temperature and automating some appliances. This is important for the energy bill and the environment, but it overlooks the most important function of our homes: providing a safe and healthy living space. Ambient plans to make a product that allows everyone in the world -- whether they heat or cool their homes -- to measure and improve the quality of air that they breathe accurately, efficiently, and affordably.

Many of us intuitively know that the right temperature can have a big impact on our productivity, but there are many other elements of buildings that are overlooked. This includes having the right level of humidity, which has been shown to reduce spread of harmful pathogens such as viruses (including COVID-19). Moreover, we are so concerned about outdoor pollution from cars and factories that we overlook the fact this same air enters our homes in various ways. At the same time, we forget about harmful pollutants emitted by common items such as wall paint, furniture finishes, and of course cook stoves. We cannot fix something that we do not see.

There are four shortfalls of current indoor air quality monitoring products: they measure only one location, they are often inaccurate, and they do not provide a way to improve air quality. The cost of these alternative varies widely from $10 to $500, making it difficult for people to choose the right option for their home.

Ambient's product solves these issues through a three-part system: a network of low-energy IoT nodes, a mobile application for network management, and a cloud-based infrastructure. This three-part system overcomes the limitations of current alternatives in three ways. First, multiple nodes can measure multiple points around the home, increasing overall accuracy of measurement thanks to additional sampling points. Second, the nodes themselves will have the bare minimum infrastructure needed for data measurement, storage, and communication. This keeps the overall cost of the system very low (less than $100). Third, the processing and analytics will be performed on the cloud and delivered to the user’s mobile device. It will include advice on improving key indoor air quality measurements along with providing real-time data.

Moreover, later versions of the product may expand to include automated adjustments to various appliances as a way to manage and improve indoor air quality. This could include fine-tuning the thermostat, humidifier, or purifier.

There are several companies that provide air quality meters, but users find them woefully inaccurate or hard to use. All of them sample only one point and are usually expensive. These include products by Temtop, Awair, and Huma-i. Then there are big tech companies like Amazon or Google who could pivot their current smart home products into this direction.

I think most companies that make smart home products do not focus on the right problems. Automating temperature and lights are important for the energy bill and the environment, but it is unfortunately not a major concern for most people, especially those who view smart home products as a luxury. What is more powerful is the need to ensure we have a safe and healthy living space. Indoor air quality is such a core component of a healthy life, which is only made clearer with the current global crisis.